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EXPERIMENT 10

Aim:

To explore the cryptocurrency landscape using real-world data and blockchain concepts.

Theory:

This study analyzes the crypto market using Chainlink (LINK) as the focus asset. Data was collected from:

- CoinGecko API – Historical price, market cap, and volume (used in Python).
- CoinMarketCap – Price charts and market ranking.
- Crypto.com – Market stats and news updates.

Cryptocurrency Landscape:

- **Definition:** Digital assets based on blockchain enabling secure, decentralized transactions.
- **Types:**
 - Store of Value – Bitcoin, Litecoin
 - Utility Tokens – Ethereum, Chainlink
 - Stablecoins – USDT, USDC
 - Governance Tokens – UNI, AAVE
- **Volatility:** Prices fluctuate due to market trends and regulations.
- **Importance:** Data analysis helps track performance and trends.
- **Advancements:** DeFi, NFTs, Layer-2 scaling, cross-chain systems.

Data Collection & Analysis:

Using CoinGecko API and Pandas:

- Daily Returns – Measures price changes
- Moving Averages (MA7, MA30) – Trend indicators
- Cumulative Returns – Overall growth
- Rolling Volatility – Risk analysis

Visualization:

A dashboard (Matplotlib) showing:

- Price Trend
- Price vs Volume
- Daily Returns Distribution
- Moving Averages
- Cumulative Returns
- Rolling Volatility

Technological Analysis:

- Consensus: PoW, PoS
- Scalability: Layer-2, sharding
- Interoperability: Chainlink CCIP, Polkadot, Cosmos
- Privacy: zk-SNARKs, ring signatures

Chainlink focuses on staking, oracle improvements, and cross-chain capabilities.

Market Trends:

- Increased crypto adoption by businesses
- Growth in crypto wallets
- Higher network activity
- Integration with fintech
- Evolving regulations

Utilizing Api for analysis and visualization : **Code:**

```
import requests

import matplotlib.pyplot as plt

import pandas as pd

# ----- Fetch Data -----

def get_historical_data(coin_id, days='180', currency='usd'):

    url = f'https://api.coingecko.com/api/v3/coins/{coin_id}/market_chart'

    params = {'vs_currency': currency, 'days': days}

    response = requests.get(url, params=params)

    return response.json()

# ----- Dashboard Function
```

```

data = get_historical_data(coin_id, days)

# Create DataFrame
prices_df = pd.DataFrame(data['prices'], columns=['timestamp', 'price'])
volumes_df = pd.DataFrame(data['total_volumes'], columns=['timestamp', 'volume'])

prices_df['timestamp'] = pd.to_datetime(prices_df['timestamp'], unit='ms')
volumes_df['timestamp'] = pd.to_datetime(volumes_df['timestamp'], unit='ms')

df = pd.merge(prices_df, volumes_df, on='timestamp')
df.set_index('timestamp', inplace=True)

# Calculate extra metrics
df['returns'] = df['price'].pct_change()
df['MA7'] = df['price'].rolling(window=7).mean()
df['MA30'] = df['price'].rolling(window=30).mean()
df['cumulative'] = (1 + df['returns']).cumprod()
df['volatility'] = df['returns'].rolling(window=7).std() * 100

# ----- Plot Dashboard -----
fig, axes = plt.subplots(3, 2, figsize=(16, 12))
fig.suptitle(f'{coin_id.capitalize()} Dashboard (Last {days} Days)', fontsize=16, fontweight='bold')

# 1. Price Trend
axes[0, 0].plot(df.index, df['price'], color='blue')
axes[0, 0].set_title('Price Trend')
axes[0, 0].set_ylabel('Price (USD)')
axes[0, 0].set_xlabel('Year-Month') # Add xlabel
axes[0, 0].grid(True)

```

2. Price vs Volume

```
axes[0, 1].plot(df.index, df['price'], color='blue', label='Price')
axes[0, 1].bar(df.index, df['volume'], color='orange', alpha=0.3, label='Volume')
axes[0, 1].set_title('Price vs Volume')
axes[0, 1].set_xlabel('Year-Month') # Add xlabel
axes[0, 1].set_ylabel('Price (USD) / Volume') # Add ylabel
axes[0, 1].legend()
axes[0, 1].grid(True)
```

3. Daily Returns Distribution

```
axes[1, 0].hist(df['returns'].dropna(), bins=50, alpha=0.7, color='purple')
axes[1, 0].set_title('Daily Returns Distribution')
axes[1, 0].set_xlabel('Daily Return')
axes[1, 0].set_ylabel('Frequency')
axes[1, 0].grid(True)
```

4. Moving Averages

```
axes[1, 1].plot(df.index, df['price'], color='lightgray', label='Price')
axes[1, 1].plot(df.index, df['MA7'], color='blue', label='7-day MA')
axes[1, 1].plot(df.index, df['MA30'], color='red', label='30-day MA')
axes[1, 1].set_title('Moving Averages')
axes[1, 1].set_xlabel('Year-Month')
axes[1, 1].set_ylabel('Price (USD)')
axes[1, 1].legend()
axes[1, 1].grid(True)
```

5. Cumulative Returns

```
axes[2, 0].plot(df.index, df['cumulative'], color='green')
axes[2, 0].set_title('Cumulative Returns')
axes[2, 0].set_ylabel('Growth Factor')
```

```
axes[2, 0].set_xlabel('Year-Month')
```

```
axes[2, 0].grid(True)
```

```
# 6. Volatility
```

```
axes[2, 1].plot(df.index, df['volatility'], color='orange')
```

```
axes[2, 1].set_title('Rolling Volatility (7-Day)')
```

```
axes[2, 1].set_ylabel('Volatility (%)')
```

```
axes[2, 1].set_xlabel('Year-Month')
```

```
axes[2, 1].grid(True)
```

```
plt.tight_layout(rect=[0, 0, 1, 0.96])
```

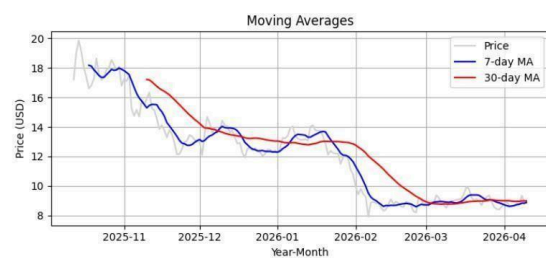
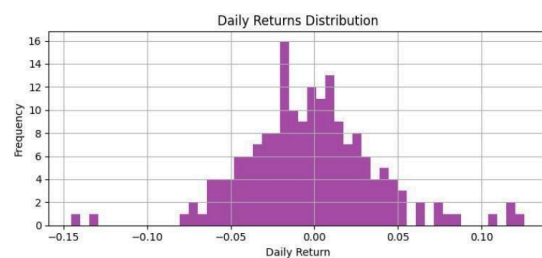
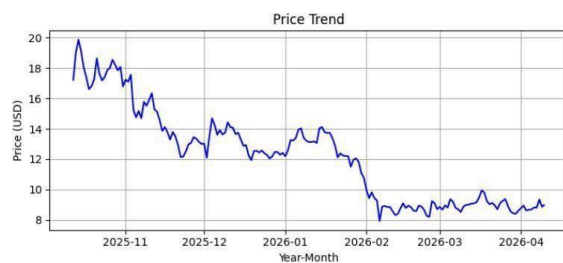
```
plt.subplots_adjust(hspace=0.5, wspace=0.3) # Adjust vertical and horizontal spacing
```

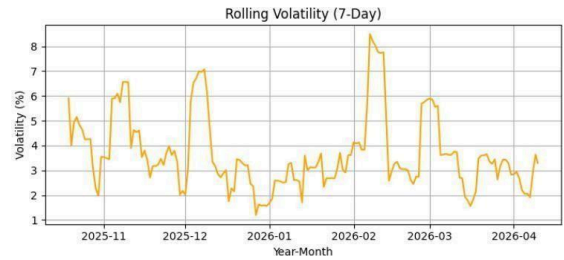
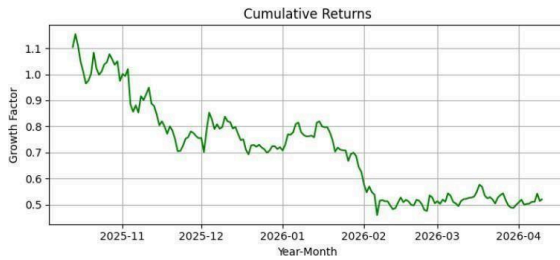
```
plt.show()
```

```
# ----- Run Dashboard
```

```
-----chainlink_dashboard('180')
```

Output:





Coingecko Website

coingecko.com

Now LIVE: Spot CEX Report 2026

Coins: 17,728 Exchanges: 1,457 Market Cap: \$2,527T ▲ 0.9% 24h Vol: \$87,101B Dominance: BTC 57.0% ETH 10.6% Gas: 0.126 GWEI

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Cryptocurrency Prices by Market Cap

The global cryptocurrency market cap today is \$2.53 Trillion, a ▲ 0.9% change in the last 24 hours. [Read more](#)

Highlights ☒

Market Cap	24h Trading Volume	Trending	Top Gainers
\$2,526,881,227,609 ▲ 0.9%	\$87,100,760,179	- Zcash: \$363.04 ▲ 11.1% - Bitcoin: \$71,924.45 ▲ 1.1% - Siren: \$0.7399 ▲ 29.8%	- RaveDAO: \$0.5855 ▲ 92.0% - Xphere: \$0.01634 ▲ 63.7% - Siren: \$0.7399 ▲ 29.8%

coingecko.com/en/coins/bitcoin

Bitcoin BTC Price #1

\$72,037.64 ▲ 1.2% (24h)

1.0000 BTC ▲ 0.0%

24h Range: \$70,522.33 - \$72,495.25

Add to Portfolio • 2,367,265 added

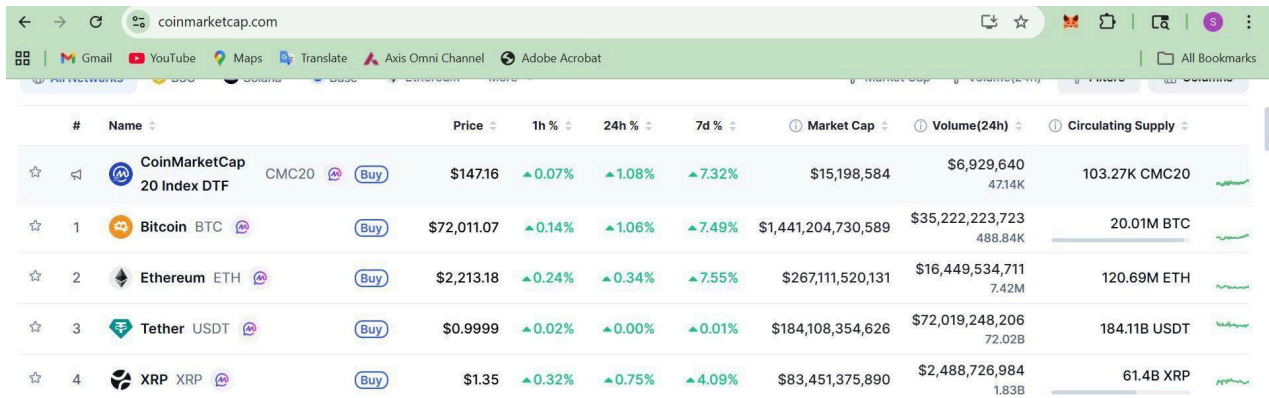
Metric	Value
Market Cap	\$1,441,734,490,650
Fully Diluted Valuation	\$1,441,734,490,650
24 Hour Trading Vol	\$35,707,938,832
Circulating Supply	20,013,637
Total Supply	20,013,637
Max Supply	21,000,000

Overview Markets Treasuries News Similar Coins Historical Data BTC Halving

Price Compare 24H 7D 1M 3M YTD 1Y Max

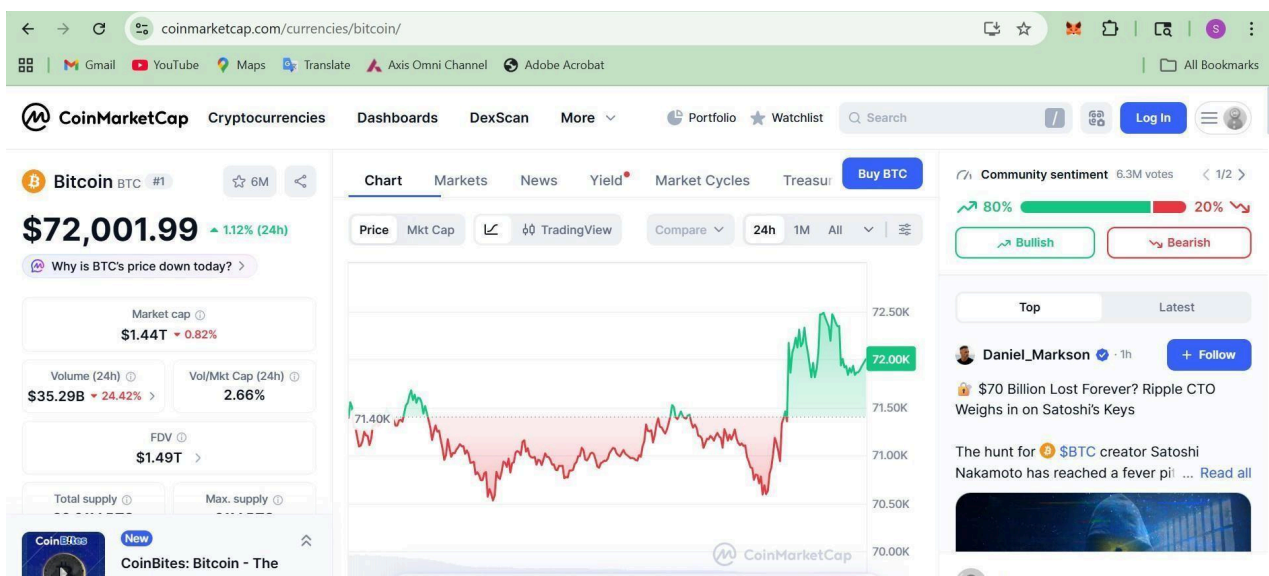
coingecko

CoinMarketCap website



This screenshot shows the top cryptocurrencies listed on the CoinMarketCap website. The table includes columns for rank, name, price, and percentage changes over 1h, 24h, and 7d. It also displays market capitalization, 24h trading volume, and circulating supply for each asset.

#	Name	Price	1h %	24h %	7d %	Market Cap	Volume(24h)	Circulating Supply
	CoinMarketCap 20 Index DTF	\$147.16	▲0.07%	▲1.08%	▲7.32%	\$15,198,584	\$6,929,640 47.14K	103.27K CMC20
1	Bitcoin BTC	\$72,011.07	▲0.14%	▲1.06%	▲7.49%	\$1,441,204,730,589	\$35,222,223,723 488.84K	20.01M BTC
2	Ethereum ETH	\$2,213.18	▲0.24%	▲0.34%	▲7.55%	\$267,111,520,131	\$16,449,534,711 7.42M	120.69M ETH
3	Tether USDT	\$0.9999	▲0.02%	▲0.00%	▲0.01%	\$184,108,354,626	\$72,019,248,206 72.02B	184.11B USDT
4	XRP XRP	\$1.35	▲0.32%	▲0.75%	▲4.09%	\$83,451,375,890	\$2,488,726,984 1.83B	61.4B XRP



Conclusion:

By combining automated market data processing, visual trend analysis, blockchain technology evaluation, and adoption metrics, this experiment presents a holistic view of the cryptocurrency ecosystem. It demonstrates how blockchain's technical foundations connect with real-world market behavior, providing insights into both current trends and future advancements in the digital asset industry.